

2012 HKDSE Physics

Paper 1B Structured Questions

Original questions with official answers and English explanations. Correct answers are red; explanations are blue.

Question 1

Cappuccino is an Italian style coffee topped with a layer of frothy milk (Figure 1.1).



Figure 1.1

Frothy milk is made by bubbling steam through milk, which is held in a metallic jug (Figure 1.2). Steam is ejected from the steam wand of a cappuccino machine (Figure 1.3).



Figure 1.2



Figure 1.3

Given: specific latent heat of vaporization of water = $2.26 \times 10^6 \text{ J kg}^{-1}$

Specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$

Specific heat capacity of steam = $2000 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$

Specific heat capacity of milk = $3900 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$

(a) Calculate the total amount of heat released when 20 g of steam at 110°C cools to 100°C and condenses to water at 100°C . (3 marks)

(b) 20 g of steam at 110°C is bubbled through 200 g of milk at 15°C to make frothy milk. Using the result in (a), estimate the temperature of the frothy milk. (2 marks)

(c) Would the actual temperature of frothy milk be higher than, equal to or lower than the value found in (b)? Explain. (2 marks)

Official marking scheme

1. (a) $Q = m_s c_s \Delta T + m_s l_v$ 1M+1M
 $= 0.02 (2000)(110 - 100) + 0.02 (2260000)$
 $= 400 + 45200$
 $= 45600 \text{ J}$ 1A 3
- (b) $m_m c_m \Delta T_m = Q + m_s c_w \Delta T_w$ 1M
 $0.2 (3900)(T - 15) = 45600 + 0.02 (4200)(100 - T)$
 $T = 76.0^\circ \text{C}$ 1A 2
- (c) The actual temperature of frothy milk is lower than the value calculated in (b). 1A
 Because energy loss by the steam is also gained by the surroundings 1A
 including the air/the metallic jug that holds the milk. 2

(a) Separate the two changes: steam first cools by 10°C ($mc\Delta T$), then condenses at constant temperature (ml). Convert 20 g to 0.020 kg. Adding the two releases gives 45.6 kJ; latent heat is not another temperature change.

(b) There is a third release: the condensed water cools from 100°C to T . Balance all released energy against heating the milk, not just the 45.6 kJ from (a). Rearranging gives $864T = 65700$, hence $T = 76.0^\circ\text{C}$. This is below 100°C , consistent with complete condensation.

(c) The ideal balance heats only the milk. In practice some energy heats the jug and air, so less heats the milk and the final temperature is lower.

Question 2

* A gas bubble rises from the bottom of a lake to the water surface. Its radius increases from 0.8 cm to 1.0 cm.

(a) If the gas pressure in the bubble at the water surface is 1.01×10^5 Pa, find the gas pressure in the bubble when it is at the bottom of the lake. Assume that the temperature of the gas in the bubble remains constant. (2 marks)

(b) Use kinetic theory to explain the change in gas pressure in the bubble as its volume increases. (2 marks)

Official marking scheme

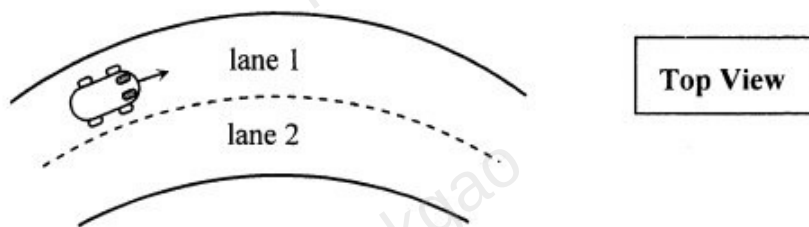
2. (a) $p_1 V_1 = p_2 V_2$ (Or $p \propto \frac{1}{V}$) 1M
- $$p_1 \left(\frac{4}{3} \pi \times (0.8)^3 \right) = (1.01 \times 10^5) \left(\frac{4}{3} \pi \times (1.0)^3 \right)$$
- $$p_1 = 1.97 \times 10^5 \text{ Pa} \quad 1A \quad \underline{2}$$
- (b) Volume increases as bubble rises but the speed / k.e. of gas molecules remains unchanged 1A
therefore frequency of collision of molecules on bubble's inner surface decreases, gas pressure decreases. 1A 2

(a) For a fixed amount of gas at constant temperature, pV is constant. The spherical volume is proportional to r^3 , not r or r^2 : $p_{\text{bottom}} = 1.01 \times 10^5 (1.0/0.8)^3 = 1.97 \times 10^5$ Pa. Both radii may use cm because only their ratio is needed.

(b) Constant temperature means unchanged average molecular kinetic energy, not slower molecules. Expansion lowers molecular number density, so fewer wall collisions occur per unit area per unit time. The momentum transfer per unit area per unit time falls, giving lower pressure.

Question 3

* Figure 3.1 shows the top view of a horizontal road with two circular lanes. A car of mass 1200 kg moves with constant speed in lane 1 of radius 45 m.



- (a)(i) Name the force that provides the centripetal force for the car. If the maximum value of this force is 8000 N, calculate the highest speed of the car such that it can keep in lane 1. (3 marks)
- (a)(ii) Suppose the car takes lane 2 instead of lane 1 and the maximum value of the force providing the centripetal force is still 8000 N. Would the car's highest speed in lane 2 be smaller than, larger than or the same as that found in (a)(i)? Explain. (2 marks)
- (b) Explain why the chance of skidding would increase if there are oil patches on the road surface in Figure 3.1. (2 marks)

Official marking scheme

3. (a) (i) Friction f between the tyres and the road.

1A

$$f = \frac{mv^2}{r}$$

1M

$$8000 = \frac{1200 v^2}{45 \text{ m}}$$

$$v = 17.3 \text{ m s}^{-1}$$

1A 3

- (ii) Smaller

1A

For the same f , $v^2 \propto r$; when r decreases v decreases.

1A 2

- (b) (Max) friction / coefficient of friction reduced,
not enough to provide the centripetal force / acceleration required for circular motion.
Or Tracking or allowed speed lowered.

1A

1A 2

(a)(i) On an unbanked horizontal road, weight and normal reaction balance vertically; tyre-road friction provides the horizontal inward resultant. Set the required mv^2/r equal to its maximum: $v_{\text{max}} = \sqrt{(8000 \times 45 / 1200)} = 17.3 \text{ m s}^{-1}$.

(a)(ii) The drawing places lane 2 on the inside, so its radius is smaller. With the same available inward force, $v_{\text{max}} \propto \sqrt{r}$ and must be smaller; the car cannot keep its old maximum speed on the tighter bend.

(b) At a given speed and radius, required mv^2/r is unchanged. Oil lowers the maximum available friction. When available friction is less than required, the car cannot follow the circular lane and skids; no extra outward force is needed to explain this.

Question 4

Train A initially travels at a speed of 60 m s^{-1} along a straight horizontal railway. Another identical train B travels ahead of A in the same direction on the same railway. Due to mechanical failure, B is only travelling at 20 m s^{-1} (Figure 4.1).

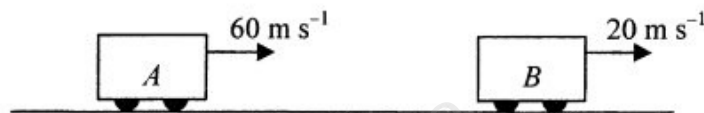
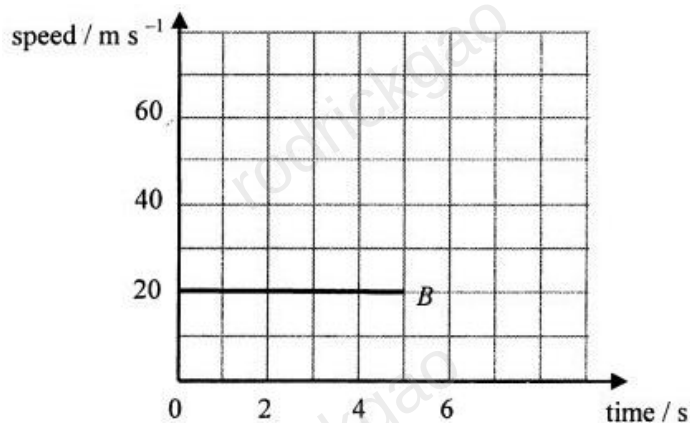


Figure 4.1

At time $t = 0$, A and B are $x \text{ m}$ apart, the captain of A receives a stopping signal and immediately A decelerates at 4 m s^{-2} while B continues to travel at 20 m s^{-1} . A eventually collides with B after 5 s. Neglect air resistance.

(a)(i) Find the speed of A just before collision. (2 marks)

(a)(ii) The graph below shows how the speed of B varies with time within this 5 s. Sketch on the same graph the variation of the speed of A within the same time period. (1 mark)



(a)(iii) Based on the above information, determine the separation x of the two trains at $t = 0$. (3 marks)

(b) A and B locked together after collision.

(b)(i) Find the speed of them just after collision. (2 marks)

(b)(ii) If the collision time between the trains is 0.2 s and the mass of each train is 5000 kg, find the magnitude and direction of the average impact force acted on A during collision. (3 marks)

Official marking scheme

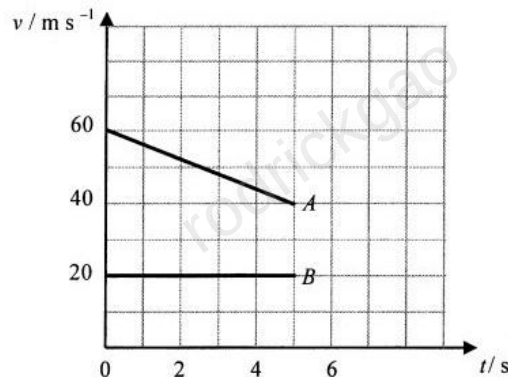
Marks

4. (a) (i) $v = u + at$
 $= 60 + (-4)5$
 $= 40 \text{ m s}^{-1}$

1M

1A 2

(ii)



1A

1

(iii) $s_A = \left(\frac{60 + 40}{2} \right)(5) = 250 \text{ m}$
 $s_B = (20)(5) = 100 \text{ m}$
 $x = 250 - 100$
 $= 150 \text{ (m)}$
 [equals to the area between the two graphs]

1M

1M

1A 3

(b) (i) $m u_A + m u_B = (m + m)V$
 $40 + 20 = 2V$
 $V = 30 \text{ m s}^{-1}$

1M

1A

2

(ii) $F = \frac{mV - mu_A}{\Delta t}$
 $= (5000) \frac{(30 - 40)}{0.2}$
 $= -250000 \text{ N}$

1M

1A

Impact force is opposite to the direction of travel of A (to the left / backwards / negative)

1A

3

(a)(i) Take forward as positive. Deceleration gives $a = -4 \text{ m s}^{-2}$, so after 5 s the speed is 40 m s^{-1} .

(a)(ii) Constant acceleration gives a straight line from $(0 \text{ s}, 60 \text{ m s}^{-1})$ to $(5 \text{ s}, 40 \text{ m s}^{-1})$; stop the line at the collision time.

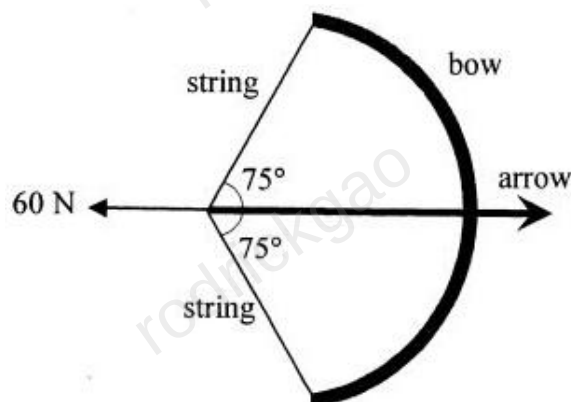
(a)(iii) A closes the initial gap by travelling farther than B, not by travelling its whole 250 m. Thus $x = s_A - s_B = 150 \text{ m}$; equivalently use the area between the speed-time lines.

(b)(i) During the brief collision, use the marking model of negligible external impulse. Momentum, not kinetic energy, is conserved. Equal masses lock together at the mean of their immediately pre-impact speeds: $(40+20)/2 = 30 \text{ m s}^{-1}$. Do not use the earlier 60 m s^{-1} .

(b)(ii) A alone loses momentum: $\Delta p = 5000(30-40) = -50000 \text{ kg m s}^{-1}$. Divide by the 0.2 s impact duration, not the 5 s braking time. Magnitude $2.50 \times 10^5 \text{ N}$, backwards.

Question 5

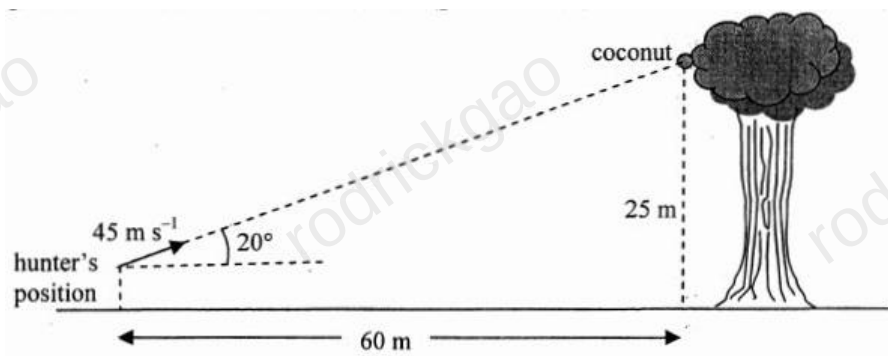
(a) A bow and arrow is a kind of projectile weapon. The string of a bow is drawn taut by a hunter with a force of 60 N and an arrow of mass 0.2 kg is held stationary as shown in Figure 5.1.



(a)(i) Find the tension of the string. Neglect the weight of the arrow. (2 marks)

(a)(ii) Estimate the energy stored in the taut string if the initial speed of the arrow is 45 m s^{-1} when released. Assume that the bow is rigid and neglect the mass of the string. (2 marks)

* (b) The hunter stands at about 60 m away from a tree as shown in Figure 5.2. He uses the bow to release the arrow in order to shoot a coconut held by a monkey (not shown in the figure) in the tree. The coconut is at a height of 25 m from the ground. The hunter aims directly at the coconut and the arrow leaves the bow at a speed of 45 m s^{-1} making an angle of 20° to the horizontal. At the moment the hunter releases the arrow, the monkey drops the coconut such that it falls vertically from rest. Neglect air resistance and the arrow's size. ($g = 9.81 \text{ m s}^{-2}$)



(b)(i) Find the time taken for the arrow to hit the coconut. (2 marks)

(b)(ii) Find the height of the coconut from the ground at the moment the arrow hits it. (2 marks)

Official marking scheme

5. (a) (i)	Let T be the tension. $2T \cos 75^\circ = 60$ $T = 115.9 \text{ N}$	1M 1A	<u>2</u>
(ii)	energy stored in the string = k.e. of arrow $= \frac{1}{2}(0.2)(45)^2$ $= 202.5 \text{ J}$	1M 1A	<u>2</u>
(b) (i)	$d = v \cos 20^\circ t$ $60 = 45 \cos 20^\circ t$ $t = 1.42 \text{ s}$	1M 1A	<u>2</u>
(ii)	$h = 25 - \frac{1}{2}gt^2$ $= 25 - \frac{1}{2}(9.81)(1.42)^2$ $= 15.1 \text{ m [or } h = 14.9 \text{ m]}$	1M 1A	<u>2</u>

(a)(i) Each sloping half of the string pulls with the same tension T . Their vertical components cancel; their two horizontal components add to balance 60 N. Hence $T = 60/(2 \cos 75^\circ) = 115.9 \text{ N}$, not 30 N.

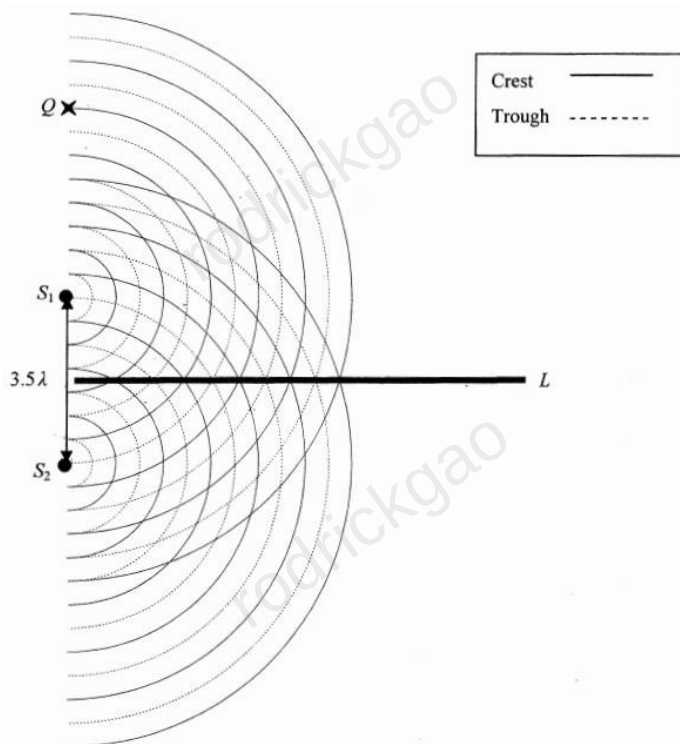
(a)(ii) Under the stated rigid-bow, negligible-string-mass model, stored energy becomes the arrow's kinetic energy: $E = \frac{1}{2}mv^2 = 202.5 \text{ J}$.

(b)(i) The coconut has no horizontal motion; the arrow reaches its horizontal position after $60/(45 \cos 20^\circ) = 1.42 \text{ s}$. Both objects acquire the same downward displacement $\frac{1}{2}gt^2$ relative to their no-gravity paths, so aiming directly at the coconut ensures interception before it reaches the ground.

(b)(ii) Use the falling coconut: its initial vertical speed is zero, so its height is $25 - \frac{1}{2}gt^2 = 15.1 \text{ m}$. This avoids wrongly treating the arrow's launch point as ground level. The answer is positive, confirming the interception occurs above the ground.

Question 6

In a ripple tank, circular water waves are produced by two vibrators S_1 and S_2 of the same frequency vibrating in phase. Their separation is 3.5λ , where λ is the wavelength of the waves. Figure 6.1 shows the two circular waves propagating on the water surface at a certain moment. Line L connects all points P which have path difference $S_1P - S_2P = 0$.



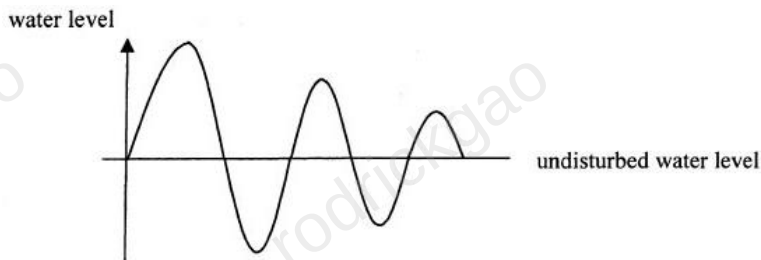
(a) Draw and label a line in Figure 6.1 connecting all points P which have path difference

(a)(i) $S_1P - S_2P = \lambda$ (label it as L_1)

(a)(ii) $S_1P - S_2P = -3\lambda/2$ (label it as L_2)

What would happen to L_1 and L_2 if the separation between S_1 and S_2 is reduced slightly? (3 marks)

(b) Figure 6.2 shows the profile of the water level along line L at a certain instant. Sketch on the same figure the profile at a time $T/2$ later, where T is the period of the water waves. (1 mark)

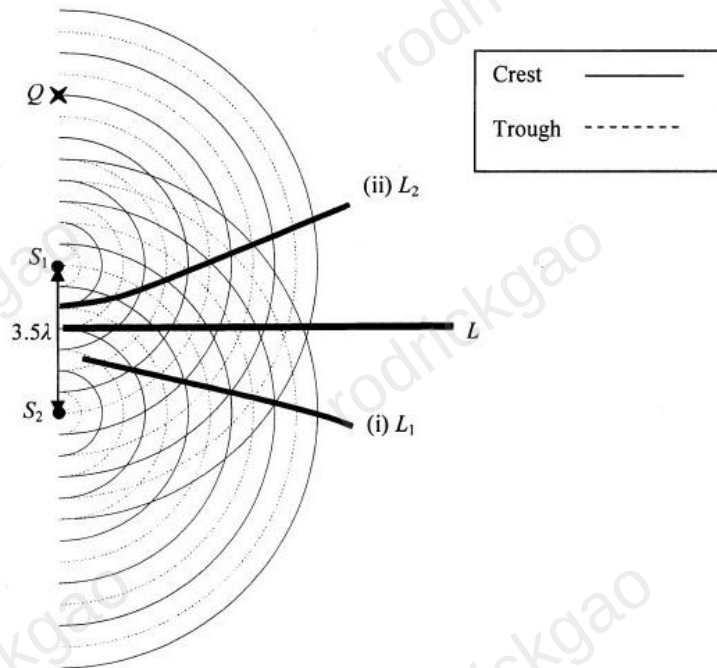


(c) Q is a point on the line joining S_1 and S_2 as shown in Figure 6.1. State the kind of interference that occurs at Q and give a reason for this occurrence. (2 marks)

*(d) A similar double-slit set-up is used for the demonstration of the interference of light in which the separation between slits S_1 and S_2 is 0.5 mm and the screen is at 2.5 m from the slits. Calculate the average separation between adjacent bright fringes on the screen for a monochromatic light of wavelength 550 nm. (2 marks)

Official marking scheme

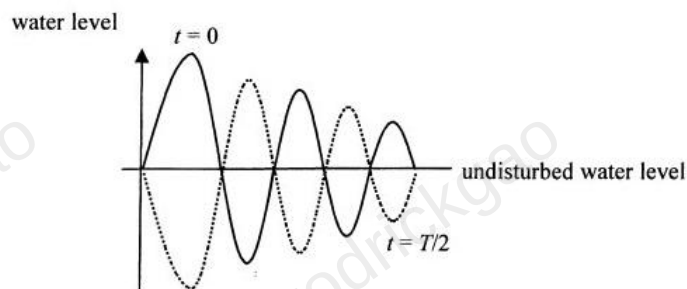
6. (a)



2A

L_1 / L_2 are further from L or separation of $L / L_1 / L_2$ is greater or angle between $L / L_1 / L_2$ larger. 1A 3

6. (b)



1A

1

- (c) The two waves are out of phase at Q as path difference $= 3.5\lambda$ ($QS_1 = 4\lambda$ and $QS_2 = 7.5\lambda$), destructive interference occurs.

1M

1A 2

(d)

$$\Delta y = \frac{D\lambda}{a} = \frac{2.5 \times 550 \times 10^{-9}}{0.5 \times 10^{-3}} = 2.75 \times 10^{-3} \text{ m}$$

1M

1A 2

(a)(i) Positive $S_1P - S_2P$ means P is farther from the upper source S_1 , so L_1 lies below L. Join positions with a one-wavelength path difference; equal phases reinforce.

(a)(ii) A negative path difference means P is closer to S_1 , so L_2 lies above L. Join positions with difference -1.5λ , where crest meets trough. Do not put both lines on the same side of L.

(a) Reducing source separation spreads a fixed-order interference pattern farther from the centre: in the far field $|\Delta| \approx d \sin|\theta|$, so fixed $|\Delta|$ and smaller d require a larger $|\theta|$.

(b) Half a period changes every local oscillation phase by π . Reverse each displacement about the undisturbed water level, keeping its local magnitude: former crests become troughs. Do not draw a flat surface merely because L is an interference maximum.

(c) Because Q lies beyond S_1 on the source line, its path-difference magnitude equals the full source separation 3.5λ . This half-integer multiple gives opposite phase and destructive interference; equal amplitudes are not required to identify the interference type.

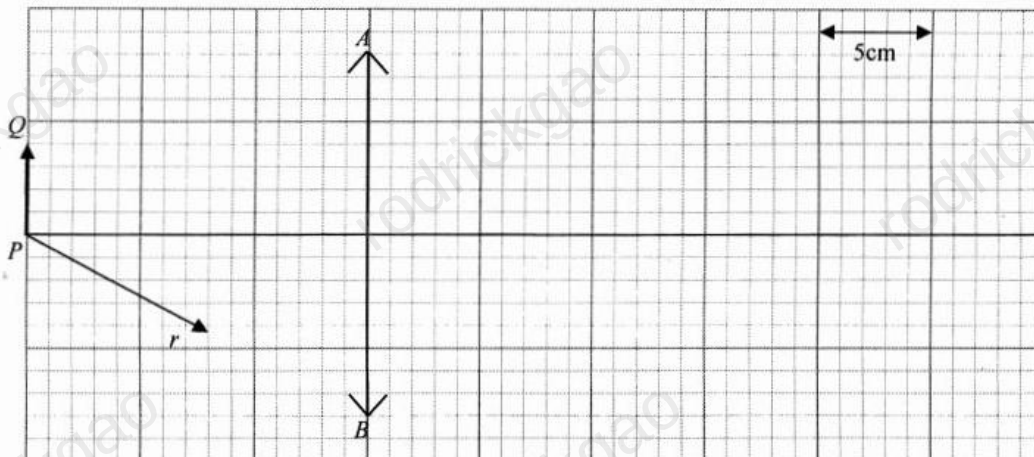
(d) Use $\Delta y = \lambda D/d$ with $550 \text{ nm} = 550 \times 10^{-9} \text{ m}$ and $0.5 \text{ mm} = 0.5 \times 10^{-3} \text{ m}$. Result $0.00275 \text{ m} = 2.75 \text{ mm}$.

Question 7

A luminous object PQ is placed 15 cm in front of a convex lens AB as shown in Figure 7.1.

(a) The focal length of the lens is 5 cm.

(a)(i) Use a graphical method to find the location of the image of the object. Clearly draw all the construction lines on Figure 7.1 and state the nature of the image. (4 marks)



(a)(ii) Complete the path of ray r on Figure 7.1 to show how it travels after passing through the convex lens. (1 mark)

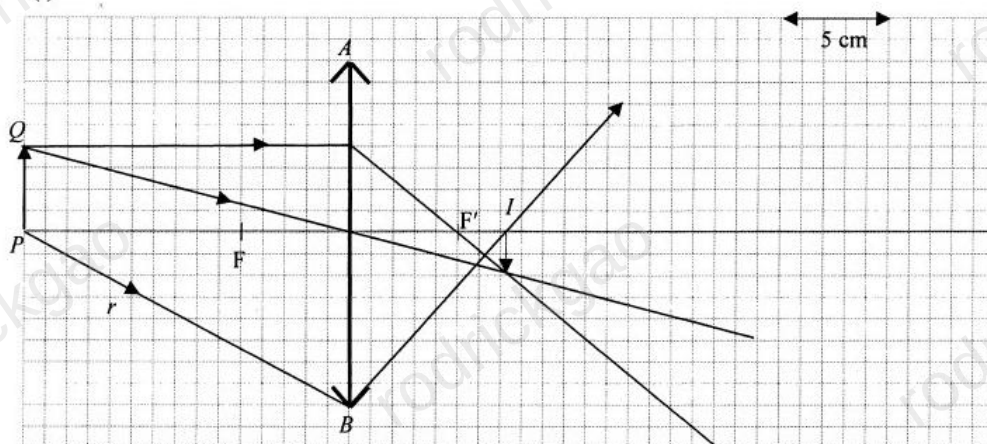
(b) Suppose that a convex lens of focal length 10 cm is used instead while the size of the lens and the object distance of PQ from the lens remain unchanged.

* (b)(i) Use the lens formula to find the image distance. Find also the linear magnification of the image. (3 marks)

(b)(ii) Compare the brightness of this image with that in (a). Explain. (2 marks)

Official marking scheme

7. (a) (i)



Two correct rays to find image I .
Nature: real, inverted, diminished.

2A
2A 4
1A 1

(ii) Ray r correctly completed.

(b) (i) $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$ $\frac{1}{15} + \frac{1}{v} = \frac{1}{10}$

$$v = 30 \text{ cm}$$

$$m = \frac{30}{15} = 2$$

1M
1A
1A 3

(ii) As the light energy collected by the lens is the same for both cases, for (b)(i), an enlarged image ($u < v$), same amount of light energy distributed over a larger image/intensity of light decreases as distance increases, i.e. dimmer image for (b)(i)

1A
1A

Or for (a), a diminished image ($u > v$), same amount of light energy distributed over a diminished image, i.e. brighter image for (a)

2

(a)(i) Mark both focal points 5 cm from the lens. From Q, draw one ray parallel to the principal axis and refract it through the far focus; draw a second ray through the optical centre undeviated. Their intersection locates the inverted tip I. Check: $v = uf/(u-f) = 7.5 \text{ cm}$ and $|m| = 0.5$, so the image is real, inverted and smaller.

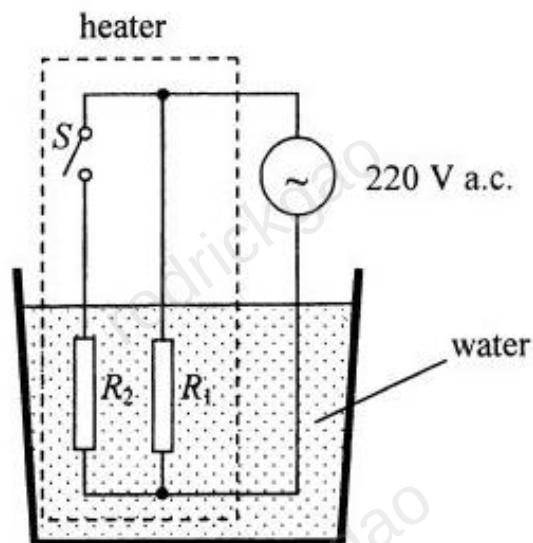
(a)(ii) Ray r originates at P, not Q. All rays from P must meet at P's image on the principal axis, 7.5 cm behind the lens. Continue r to the lens and then through that axis point; do not direct it towards the inverted tip I.

(b)(i) With $f = 10 \text{ cm}$ and $u = 15 \text{ cm}$, $1/v = 1/10 - 1/15 = 1/30$. Thus $v = 30 \text{ cm}$ and the linear magnification magnitude is $v/u = 2$.

(b)(ii) Lens aperture and object distance stay fixed, so the collected light is unchanged in the ideal model. The new image is larger, spreading that light over a larger area, hence dimmer. The reason is not that a longer focal length automatically makes the lens absorb more light.

Question 8

In the circuit shown in Figure 8.1, resistors R_1 and R_2 represent the heating elements in a heater using mains supply. Both resistors are immersed in water.



The heater can be operated in two modes, namely, heating and keeping warm, and it is controlled by the switch S. The power consumed by the heater in the heating mode is 550 W and in the mode of keeping warm is 88 W. The mains voltage is 220 V a.c.

- In which mode is the heater operating when switch S is open? (1 mark)
- Find the resistance of R_1 . (2 marks)
- When switch S is closed, calculate the current passing through resistor R_2 . (3 marks)
- *What is the peak value of the sinusoidal current flowing through the heater when switch S is closed? (2 marks)

Official marking scheme

8. (a) Keeping warm / 88 W

1A 1

$$(b) R_1 = \frac{V^2}{P} = \frac{220^2}{88} = 550 \, \Omega$$

1M

1A 2

$$(c) \text{ Total current } I_0 = \frac{P_0}{V} = \frac{550}{220} = 2.5 \, \text{A}$$

$$\text{Current in } R_1, I_1 = \frac{220}{550} = 0.4 \, \text{A}$$

$$\text{Current in } R_2, I_2 = 2.5 - 0.4 = 2.1 \, \text{A}$$

Or	Power to R_2 , $550 \, \text{W} - 88 \, \text{W} = 462 \, \text{W}$
	Current in R_2 , $I_2 = \frac{P_2}{V} = \frac{462}{220} = 2.1 \, \text{A}$

1M

1M

1A 3

$$(d) \text{ Peak current} = \sqrt{2} (2.5 \, \text{A}) = 3.54 \, \text{A}$$

1M

1A 2

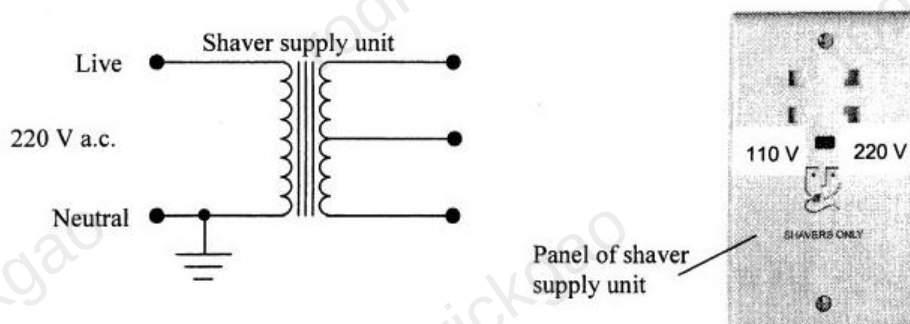
- (a) Opening S disconnects only R_2 . R_1 still heats, but the total power is smaller: keeping warm.
- (b) In that mode only R_1 receives the full 220 V, so $P = V^2/R$ gives $R_1 = 550 \, \Omega$. The 550 W heating-mode power is not the power of R_1 alone.
- (c) Closing S adds R_2 in parallel, not in series. R_1 still consumes 88 W at 220 V, leaving $550 - 88 = 462$ W for R_2 . Thus $I_{2,\text{rms}} = 462/220 = 2.1$ A.
- (d) The question asks for the entire heater current, so use $I_{\text{rms}} = 550/220 = 2.5$ A, not the 2.1 A branch current. For a sine wave $I_{\text{peak}} = \sqrt{2} I_{\text{rms}} = 3.54$ A.

Question 9

Read the following description about the “shaver supply unit” in bathrooms and answer the questions that follow.

The danger of electric shock is particularly high in bathrooms. Normal electric socket outlets should not be installed in bathrooms. As electric shavers and toothbrushes are becoming popular these days, a special unit, called “shaver supply unit”, is now common in bathrooms to provide electricity just for these low power consumption electric appliances (Figure 9.1).

The shaver supply unit consists of a transformer in which the secondary is not earthed and is completely isolated from the 220 V a.c. mains supply connecting to the primary. It can be used with 220 V or 110 V shavers.



- (a) Explain why the chance of electric shock is high in bathrooms. (2 marks)
- (b) Explain what would happen if the human body touches
 - (b)(i) the live wire of the mains supply in the primary circuit; (2 marks)
 - (b)(ii) one of the conducting wires in the shaver circuit outlet. (2 marks)
- * (c) What is the turns ratio of the primary coil to the secondary coil of the transformer so as to provide 110 V? (1 mark)

Official marking scheme

- | | | | |
|----------|--|----------|----------|
| 9. (a) | Humid conditions in bathrooms and water is a conductor which provides a conducting path or lowers the resistance between our hands/body and the source of electricity. | 1A
1A | <u>2</u> |
| (b) (i) | The person will get electric shock because full potential drop 220 V is applied to or substantial/large current flows through the human body. | 1A
1A | <u>2</u> |
| (b) (ii) | The person will not get electric shock / Nothing happens because the current in secondary circuit has no return path/incomplete circuit. | 1A
1A | <u>2</u> |
| (c) | Primary : Secondary = 2:1 for 110 V | 1A | <u>1</u> |

- (a) The key chain is wet skin/surfaces $\rightarrow$ lower resistance $\rightarrow$ larger current for the same voltage ($I = V/R$).
“There is water” alone does not explain the increased risk.
- (b)(i) In the stated mains arrangement, neutral is earthed. A body connected to earth and touching live can complete a return path, placing a large voltage across the body.
- (b)(ii) The marking answer assumes an ideal, unearthed, isolated secondary and contact with only one output conductor. Earth does not provide a return connection to the other secondary terminal, so no closed conducting loop forms. This does not make two-wire contact or faulty/leaky real equipment safe; do not test it by touching conductors.
- (c) For the ideal transformer, $N_p/N_s = V_p/V_s = 220/110 = 2$. The requested order is primary : secondary = 2 : 1.

Question 10

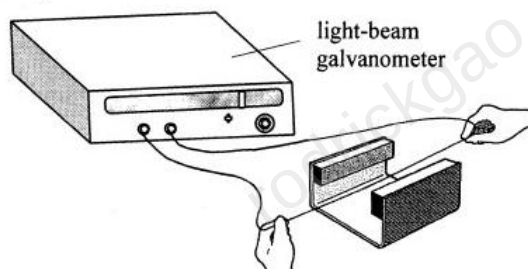
You are given a long conducting wire, a pair of slab-shaped magnets on steel yoke and a light-beam galvanometer for detecting small currents. With the aid of a diagram, describe an experiment to investigate TWO factors affecting the e.m.f. induced in a conductor when it moves in a magnetic field. (7 marks)



Official marking scheme

10. Diagram for set-up:

1A



1A
1A

Connect the long wire to the galvanometer./Connect the apparatus as shown.
Put/Place/Move the wire across the magnetic field between the pair of magnets.

Relative movement: Move the wire vertically down across the field and then up across the field, the (light-spot of the) galvanometer would deflect to one side and then to the opposite side.

Polarities of magnet: Move the wire vertically down across the field, the (light-spot of the) galvanometer would deflect to one side and then reverse the polarities of the magnet to repeat the experiment, the galvanometer would deflect to the opposite side.

Direction of movement: Move the wire vertically up/down across the field, the (light-spot of the) galvanometer would have large deflection. No deflection when the wire is moved horizontally to the left/right in the field instead.

Orientation of conductor: With the wire perpendicular to the magnetic field, move it vertically up/down across the field, the (light-spot of the) galvanometer would have large deflection. No deflection when the wire is placed parallel to the field instead.

Rate of movement: Move the wire slowly across the field and then quickly across the field, the (light-spot of the) galvanometer would deflect more for a quicker movement or a faster rate.

No. of turns: Wind the wire into, say, a 10-turn coil and move it again across the field, the (light-spot of the) galvanometer would deflect more for more turns of wire.

ANY
TWO
@2A

7

Set-up: join both ends of the same long wire to the galvanometer, with a straight part in the magnet gap perpendicular to the field. Use the official set-up diagram above as the labelled arrangement. A closed circuit is needed for the galvanometer to detect induced current.

Factor 1 — speed: keep the magnets, active wire length, wire orientation and circuit unchanged. Move the wire through the same path slowly, then faster. Larger deflection for the faster crossing shows a larger induced e.m.f.; with these perpendicular directions, $|\mathcal{E}| = Blv$.

Factor 2 — magnetic polarity: keep the wire path and speed the same but reverse the two magnet poles. The deflection reverses, showing reversal of the induced e.m.f. polarity. This is a direction test, not evidence that the magnitude increased.

Keep circuit resistance and meter setting fixed so differences in current deflection can be attributed to induced e.m.f. Give both the controlled change and the observation for each factor; merely naming “speed” and “magnetic field” is insufficient.

Question 11

Radium-226 ($^{226}_{88}\text{Ra}$) undergoes α -decay into radon (Rn).

(a) Write a nuclear equation for the decay. (2 marks)

*(b) Given: mass of a radium nucleus = 226.0254 u

Mass of a radon nucleus = 222.0176 u

Mass of an α -particle = 4.0026 u

Calculate the energy released in the decay in MeV. (2 marks)

(c) 1 curie (Ci) is defined as the activity of 1 g of radium. The activity of a radium source used in laboratories is about 5 μCi . Estimate the number of radium atoms in this source and hence find its activity expressed in disintegrations per second. The half-life of radium-226 is 1600 years and take the mass of one mole of radium as 226 g. (1 $\mu\text{Ci} = 1 \times 10^{-6} \text{ Ci}$) (3 marks)

Official marking scheme

11. (a) $^{226}_{88}\text{Ra} \rightarrow ^{222}_{86}\text{Rn} + ^4_2\text{He} / \text{He} / \alpha$

2A 2

(b) $\Delta m = 226.0254 - (222.0176 + 4.0026) = 0.0052 \text{ u}$

1M

Energy released = $(0.0052)(931) = 4.84 \text{ (MeV)}$

1A 2

(c) Number of radium atoms in the source

$$N = N_A \left(\frac{1}{226} \right) \times (5 \times 10^{-6}) = (6.02 \times 10^{23}) \frac{1}{226} \times (5 \times 10^{-6}) = 1.33 \times 10^{16}$$

1A

$$\text{Activity } A = \frac{\ln 2}{t_{1/2}} \cdot N$$

1M

$$= \frac{\ln 2}{1600 \times 365 \times 24 \times 3600} \cdot 1.33 \times 10^{16}$$

$$= 1.83 \times 10^5 \text{ (disintegrations per second, Bq)}$$

1A 3

(a) An α -particle carries mass number 4 and atomic number 2. Conserve both totals: $226 = 222 + 4$ and $88 = 86 + 2$, giving radon-222 and a helium-4 nucleus.

(b) Use the given nuclear masses consistently. The positive mass loss is 0.0052 u; $1 \text{ u } c^2 = 931 \text{ MeV}$ gives 4.84 MeV. This is the total released energy, not necessarily the kinetic energy of the α -particle alone because the daughter nucleus recoils.

(c) For the same isotope, activity is proportional to atom number and mass. Under the question's definition, 5 μCi corresponds to $5 \times 10^{-6} \text{ g}$ of radium. Divide by 226 g mol^{-1} and multiply by N_A to obtain 1.33×10^{16} atoms.

Then apply $A = \lambda N$ with $\lambda = \ln 2 / T_{1/2}$. Convert 1600 years to seconds first ($5.04576 \times 10^{10} \text{ s}$); this gives about $1.83 \times 10^5 \text{ Bq}$. Do not treat μCi as a mass unit without first using the definition in the question.